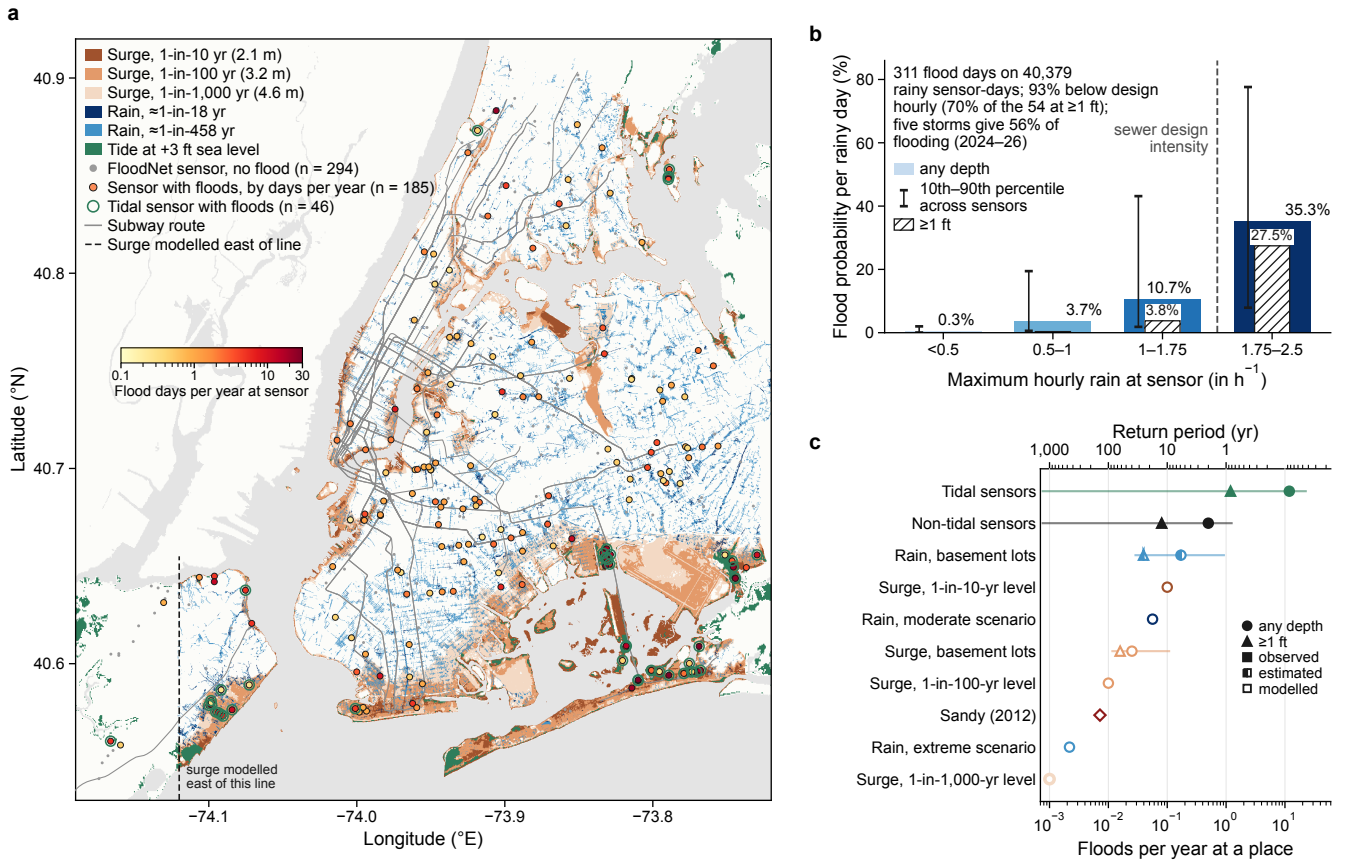

a
Surge, 1-in-10 yr (2.1 m)
Surge, 1-in-100 yr (3.2 m)
Surge, 1-in-1,000 yr (4.6 m)
Rain, ≈1-in-18 yr
Rain, ≈1-in-458 yr
Tide at +3 ft sea level
FloodNet sensor, no flood (n = 294)
Sensor with floods, by days per year (n = 185)
Tidal sensor with floods (n = 46)
Subway route
Surge modelled east of line
0.1
1
10
30
Flood days per year at sensor
surge modelled
east of this line
40.9
40.8
40.7
40.6
Latitude (°N)
−74.1
−74.0
−73.9
−73.8
Longitude (°E)
b
311 flood days on 40,379
rainy sensor-days; 93% below design
hourly (70% of the 54 at ≥1 ft);
five storms give 56% of
flooding (2024–26)
sewer design
intensity
any depth
10th–90th percentile
across sensors
≥1 ft
Flood probability per rainy day (%)
0
20
40
60
80
0.3%
3.7%
3.8%
10.7%
27.5%
35.3%
<0.5
0.5–1
1–1.75
1.75–2.5
Maximum hourly rain at sensor (in h−1)
c
Return period (yr)
1,000
100
10
1
Tidal sensors
Non-tidal sensors
Rain, basement lots
Surge, 1-in-10-yr level
Rain, moderate scenario
Surge, basement lots
Surge, 1-in-100-yr level
Sandy (2012)
Rain, extreme scenario
Surge, 1-in-1,000-yr level
any depth
≥1 ft
observed
estimated
modelled
10−3
10−2
10−1
100
101
Floods per year at a place